Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India
(Autonomous College of Affiliated to University of Mumbai)

End Semester Examination
December 2023

Maxi Marks: 100 Duration: 3 hours
Class: T.E Semester: V
Course code: CS301 Branch: COMP/CSE-AIML/CSE-DS
Name of the course: Theory of Computation

Instructions:
(1) All questions are compulsory
(2) Partial credit will be given, so show your work where appropriate.
(3) Note: DFA = Deterministic Finite Automata, NFA = Nondeterministic Finite Automata, PDA = Push-Down Automata, CFG=Context Free Grammar, TM = Turing Machine.
(4) We say that a language $L \subseteq \Sigma^*$ is reducible to language $L' \subseteq \Gamma^*$ which we denote by $L \leq L'$, if there is a computable function $f: \Sigma^* \rightarrow \Gamma^*$, where for every w, $w \in L \Leftrightarrow f(w) \in L'$.

Q No		Max Marks	CO	BL
Q1 A	A vending machine is an automatic vending machine that gives snacks, beverage etc after inserting some money into it. vending machine receives a coin of 1, 2, and 5 rupees and gives some snacks after the total amount of money equal to or greater than 5. The machine does not return money in case higher amount of money is inserted into the system. Design a deterministic finite automaton for the given scenario of vending machine.	8	1	4
B	Design a Moore machine that count the occurrence of 101 in a binary string. (Number of times 101 is available in the string)	7	1	3
C	Using pumping lemma prove that following language is not regular. $L = \{a^n b^l a^k \mid n = l \text{ or } l \neq k\}$	5	2	3
Q2 A	What is Chomsky hierarchy is all about? where is the place for recursive enumerable and recursive language in the hierarchy?	8	3	3
B	Reduce the following production of grammar to Greibach Normal Form. $A1 \rightarrow A2A1 \mid a$ $A2 \rightarrow A1A2 \mid b$	7	3	3
C	Derive the given sentences using left most derivation and right most derivation. Draw the parse tree for the derivation process. G: $S \rightarrow aB \mid bA$ $A \rightarrow a \mid aS \mid bAA$ $B \rightarrow b \mid bS \mid aBB$ 1. aabb 2. baab **OR** What do you mean by ambiguous grammar? Prove that the following grammar is ambiguous. $S \rightarrow aS \mid aSbs \mid \varepsilon$	5	3	3

Q3 A	Write regular expressions for the following languages. 1. The set of strings of 0's and 1's whose tenth symbol from the right end is 1. 2. The set of strings of 0's and 1's with at most one pair of consecutive 1's.	5	2	3
B	Construct the ε NFA for the following regular expressions: 1. RE = (a* + b*)* 2. RE = (01*0 + 10*1)*	5	1	3
Q4 A	Convert the following grammar into Chomsky Normal form (S is the start variable): $S \rightarrow ASB \mid \varepsilon$ $A \rightarrow aAS \mid a$ $B \rightarrow SBS \mid A \mid bb$	8	3	3
B	Give equivalent NPDA for the following grammar (S is the start variable): $S \rightarrow aSbb \mid T$ $T \rightarrow bTaa \mid S \mid \varepsilon$	5	4	3
C	Give Turing machines (with informal as well as formal descriptions) for the following language: Given a word w ∈ {a,b }* on the first tape, keep only every alternate letter of w on that tape. That is, suppose abbab is written on the tape, you should finally end up with abb on the same tape. You may design a 2-tape TM for this.	7	4	3
Q5 A	Prove or disprove the following statement: Let L and L' be languages such that L is reducible to L' and L' is decidable then L is also decidable.	5	4	3
B	Prove or disprove the following statement: If L is Turing recognizable and $L \leq \bar{L}$ then L is decidable.	5	4	4
Q6 A	Let <M> denote the description of a Turing machine M. Let L1 = {<M> | M halts on input <M>}. Is L1 decidable? Is L1 recursively enumerable? Justify your answers.	8	4	4
B	Construct a CFG which can generate the following language. L = {w ε {a,b}* | w has an even number of a's} **OR** Construct a CFG which generates the language consisting of binary strings which start with a 1 and do not contain 001 as a substring	7	3	3
C	State True or False with justification. Let L1 and L2 be two languages such that L1 polynomial time reduces to L2. If L2 is NP-hard, then L1 is also NP-hard.	5	4	2